NNN-B-789
March 25, 1963
SUPERSEDING
Fed. Spec. DD-V-581a (in Part)
September 22, 1953
(See section 6)

FEDERAL SPECIFICATION

BURET, STRAIGHT, PRECISION

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers glass burets suitable for laboratory purposes.

1.2 **Classification.**

1.2.1 *Types, styles, classes, grades and sizes.* Burets covered by this specification shall be of the following types, styles, classes, grades, and sizes, as specified (see 6.2).

.2.1.1 *Types.*

Type I—Borosilicate glass

Type II—Soda-lime glass

1.2.1.2 *Styles.*

Style I—Straight

Style II—Straight, 3-way stopcock

Style III—Straight, 3-way stopcock, automatic zero

1.2.1.3 *Classes and grades.*

Class A—Precision

Grade 1—National Bureau of Standards (NBS) certificate required

Grade 2—NBS certificate not required.

Class B—General purpose.

1.2.1.4 *Sizes.* Burets shall be of the following sizes in the styles and classes shown.

	Applicable styles and classes
Size 1—5 milliliters (ml), (0.01 ml divisions).	Style I, class A; style II, class A.
Size 2—5 ml (0.02 ml divisions).	Style I, class B; style II, class B.
Size 3—10 ml (0.02 ml divisions).	Style I, class A; style II, class A.
Size 4—10 ml (0.05 ml divisions).	Style I, classes A and B; style II, classes A and B.
Size 5—25 ml.	All styles and both classes.
Size 6—50 ml.	All styles and both classes.
Size 7—100 ml.	All styles and both classes.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

DD-G-541—Glass (Laboratory).

UU-T-111—Tape; Paper, Gummed (Sealing and Securing).

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

PPP-B-566—Boxes, Folding, Paperboard.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636—Box, Fiberboard.

PPP-B-676—Boxes, Set-Up, Paperboard.

PPP-C-843—Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

Fed. Test Method Std. No. 626—Glassware; Methods for Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specification:

MIL-M-14077—Molding Plastic, Polytetrafluoroethylene (TFE-Fluoro-Carbon Resin).

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Uniform Classification Committee Publication:

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, Chicago 6, Illinois.)

National Bureau of Standards

NBS Circular 602 — Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

Commercial Standard:

CS21—Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Borosilicate glass.* Borosilicate glass for type I burets shall conform to the glass requirements of type I, class A of DD-G-541.

3.1.2 *Soda-lime glass.* Soda-lime glass for type II burets shall conform to the glass requirements of type II of DD-G-541.

3.1.3 *Polytetrafluoroethylene.* Polytetrafluoroethylene shall conform to the composition requirements of class 1 of MIL-M-14077.

3.2 Annealing. Ring strain shall be no greater than that of the standard reference disc No. 1 when tested as specified in 4.6.1. Any longitudinal strain, if present, shall be faint and highly diffused.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

3.3 **Capacity accuracy.**

3.3.1 *Class A.* Class A burets shall have a capacity accuracy in accordance with the requirements of NBS Circular 602.

3.3.1.1 *Grade* 1. Class A, grade 1 burets shall be marked by the manufacturer with an identification number. Each buret shall be passed by the NBS, be etched with the NBS official precision stamp, and shall have an NBS certificate of capacity. No further testing for capacity accuracy shall be required.

3.3.1.2 *Grade* 2. Class A, grade 2 burets may, but need not be marked by the manufacturer with an identification number. Capacity accuracy for grade 2 burets shall be the same as for grade 1 burets when tested as specified in 4.6.2, but shall not be required to be submitted for NBS testing.

3.3.2 *Class B.* Class B burets shall have a capacity accuracy in accordance with the plus or minus (±) tolerances of table I when tested as specified in 4.6.2.

3.4 **Design.**

3.4.1 *Shape.* Any cross section of the graduated portion of the buret taken in a plane perpendicular to its longitudinal axis shall be circular. The shape shall permit complete emptying and thorough cleaning.

3.4.2 *Stopcocks.* All burets shall have permanently attached stopcocks. Stopcock plugs shall be provided with a retaining device. Stopcocks shall not leak when tested as specified in 4.6.3. Class A, grade 1 burets shall have stopcocks with glass plugs. Unless otherwise specified, class A, grade 2 and class B burets shall have stopcocks with glass plugs (see 6.2). All glass stopcocks shall conform to Commercial Standard CS21. If specified, class A, grade 2 and class B burets shall have stopcocks with polytetrafluoroethylene plugs (see 6.2). The polytetrafluoroethylene plugs may be of conventional or so-called reverse taper design.

3.4.3 *Buret tips.* Buret tips shall be permanently attached to the buret, shall be made with a gradual taper of 2 to 3 centimeters (cm), and the end of the tip shall be perpendicular to the longitudinal axis of the tip. The outside edge of the tip shall be beveled slightly and the end and the bevel shall be polished. Sudden constriction at the orifice shall not be acceptable.

3.4.4 *Dimensions.* Class A burets shall comply with NBS Circular 602 for dimensions and dimensional tolerances. Class B burets' dimensions and dimensional tolerances shall be as shown in table I. All burets shall be calibrated "to deliver" (symbol TD) at 20°C. Graduation markings shall extend to not less than 2.5 cm from the permanently attached stopcocks.

3.4.5 *Graduation marking.* Graduation markings shall be sharply defined lines of uniform width (maximum of 0.3 mm). Graduation markings shall be perpendicular to the vertical axis of the buret and parallel

TABLE I. *Dimensions and tolerances for class B burets*

Size	Capacity, ml.	Minimum wall thickness, millimeters (mm.)	Least graduation, ml.	Scale length, cm.	Minimum outflow-time,* seconds	Plus or minus tolerance, ml.
2	5	1.25	0.02	30 ± 5	40	0.02
4	10	1.15	0.05	35 ± 10	25	0.04
5	25	1.1	0.1	35 ± 10	30	0.06
6	50	1.2	0.1	55 ± 5	50	0.10
7	100	1.25	0.2	60 ± 5	90	0.20

* Maximum outflow-time shall be 3 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

to each other. Graduation markings shall have a clear space of at least 1 mm between two adjacent marks. The value of all main graduation markings shall be suitably inscribed in Arabic numerals directly above the line referenced. There shall be no appreciable change in the appearance of markings (3.4.5.1.2, 3.4.5.2, and 3.5) when tested as specified in 4.6.4.

3.4.5.1 *Class A.*

3.4.5.1.1 *Grade* 1. Graduation markings on class A, grade 1 burets shall conform to NBS Circular 602.

3.4.5.1.2 *Grade* 2. Graduation markings on class A, grade 2 burets shall be applied by one of the following methods—etched and filled with a permanent pigment or by application of a stain which is fired into the glass without etching. Least graduation markings shall extend a minimum of 1/2 and intermediate graduation markings shall extend a minimum of 2/3, respectively, the way around the outside circumference of the buret. If the graduation markings are etched, the main graduation markings shall extend completely around the outside circumference of the buret; if the graduation markings are stained without etching, the main graduation markings shall extend a minimum of 3/4 the way around the outside circumference of the buret.

3.4.5.2 *Class B.* Graduation markings on class B burets shall be applied by one of the following methods—etched and filled with a permanent pigment, by application of a stain which is fired into the glass without etching, or by application of an enamel which is fused on the glass without etching. Least graduation markings shall extend a minimum of 1/6 and intermediate graduation markings shall extend a minimum of 1/4, respectively, the way around the outside circumference of the buret. If the graduation markings are etched, the main graduation markings shall extend completely around the outside circumference of the buret; if the graduation markings are stained or enameled without etching, the main graduation markings shall extend a minimum of 3/4 the way around the outside circumference of the buret.

3.4.5.3 *Sizes* 1 *to* 3. For style I and style II burets, the graduation markings of sizes 1, 2, and 3 shall start not less than 3 cm or more than 7.5 cm from the upper end.

3.4.5.4 *Sizes* 4 *to* 7. For style I and style II burets, the graduation markings of sizes 4, 5, 6, and 7 shall start not less than 4 cm or more than 7.5 cm from the upper end.

3.5 **Identification marking.** All burets shall be permanently and legibly marked with the manufacturer's name or trademark, its capacity, the symbol "TD", and the temperature of calibration (20°C). Class A, grade 1 burets shall be further marked in accordance with NBS Circular 602.

3.6 **Workmanship.** All burets shall be transparent and free from chips, cracks, and other defects which impair their serviceability or distort the appearance of the liquid surface.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Special provisions.**

Source: https://assist.dla.mil -- Downloaded: 202610-26T12:48Z
Check the source to verify that this is the current version before use.

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedure which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of 3.1.1, 3.1.2, 3.1.3, 3.4.2, and section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the burets of the same type, same style, same class, same grade, and same size produced by one manufacturer for delivery at one time. Class A, grade 2, and class B buret lots shall be further defined by the type of stopcock plug required (see 6.2).

4.4 **Sampling.**

4.4.1 *For nondestructive examination.* Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2 *For test.* A sample of 3 burets shall be taken at random from each 100 burets or fraction thereof in the lot. Each buret in the sample shall be tested for annealing, leakage, and capacity accuracy (in that order) prior to permanency of marking test. The leakage and capacity accuracy tests may be done concurrently. A different 1/3 of the sample shall be used for test as specified in 4.6.4.1, 4.6.4.2.1, and 4.6.4.2.2.

4.5 **Examination.** Samples of burets and level A and level B preparation for delivery thereof shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.5.1 *Classification of defects.*

4.5.1.1 *Burets.*

Categories	Defects	Acceptance standards
**Critical:*		
1	Class A, grade 1 buret not passed by NBS	3.3.1.1
2	NBS official precision stamp or certificate missing from class A, grade 1 buret.	3.3.1.1
Major:	*AQL 2.5 percent defective*	
101	Shape not as specified	
102	Stopcock not as specified	
103	Delivery tip not as specified	
104	Temperature of calibration missing or not as specified	
105	Buret not calibrated "to deliver"	
106	Dimensions or graduation markings not as specified	
107	Buret damaged (chips, cracks, or other defects or distortion)	
108	Identification marking illegible, incorrect, or incomplete	

* Every buret in each lot of class A, grade 1 burets shall be inspected for these defects.

4.5.1.2 *Level A and level B preparation for delivery (section 5).*

Categories	Defects
Critical:	None defined
Major:	*AQL 1.5 percent defective*
101	Unit, intermediate, or shipping container not as specified.
102	Number of burets per intermediate or shipping container not as specified or indicated.
103	Unlike burets packaged per intermediate, or shipping container not as specified.
104	Closure of unit, intermediate, or shipping container not as specified.
105	Unit, intermediate, or shipping container damaged.
106	Cushioning material not as specified.
107	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.6 **Tests.** Distilled water and reagent grade chemicals shall be used throughout the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.6.1 *Annealing.* Burets shall be tested in accordance with method No. 420 of Fed. Test Method Std. No. 626.

4.6.2 *Capacity accuracy.* Class A, grade 2 and class B burets shall be tested for capacity accuracy in accordance with the test procedure of section 3.5 of NBS Circular 602.

4.6.3 *Stopcock leakage.* The buret shall be tested for leakage with the stopcock free from grease, the barrel and plug wetted with water, and the buret filled initially to the zero line with water. The rate of leakage, with the plug in either of the fully closed positions shall not exceed the following:

Nominal Capacity ml.	*Leakage ml./min.*
5	0.002
10	0.003
25	0.004
50	0.005
100	0.006

Three-way stopcocks, in addition to complying with this test, shall not show a rate of leakage greater than that given in the above table when tested similarly, with the buret empty, the plug in the normal delivery position, and the filling tube connected to a suitable graduated tube filled with water to a level 25 cm above the zero line of the buret.

4.6.4 *Permanency of marking.* A different buret shall be used for each of the following tests:

4.6.4.1 *Resistance to alkali.* The buret, with stopcock removed, shall be completely immersed in 1*N* sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.6.4.2 *Resistance to acid.*

4.6.4.2.1 *Hydrochloric acid.* The buret shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the acid for an additional 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.6.4.2.2 *Sulfuric acid.* The buret shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand 1 hour. The buret shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.7 **Acceptance/rejection criteria.** If any buret fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Fed. Std. No. 102.

5.1 **Packaging.** Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Each buret shall be packaged in a box of appropriate size conforming to PPP-B-566, style I or II, type and class optional or PPP-B-676, type, class, and style optional. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to prevent shifting of contents. Box closure shall be secured with tape conforming to UU-T-111.

5.1.1.2 *Intermediate packaging.* The number of burets specified (see 6.2), of the same type, style, class, grade, and size, pack-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

aged in accordance with 5.1.1.1, shall be intermediately packaged in a snug-fitting fiber box conforming to type I, class 1 of PPP-B-636. Closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* Burets shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Burets shall be packed in a box conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification or appendix thereto. Containers shall be closed and strapped in accordance with the applicable box specification and appendix thereto.

5.2.2 *Level B.* Burets shall be packed in fiberboard boxes conforming to PPP-B-636, type I or II, class 1, style RSC, minimum bursting strength 275 pounds per square inch.

5.2.3 *Level C.* Burets shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** Burets covered by this specification are intended for laboratory applications. In general, class B burets will be most applicable for Federal and Military agencies. Grade 1 and grade 2 burets of class A are intended for precision measurements and as calibrated reference burets.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Level of packaging and packing required.

(*c*) Type, style, class, grade, and size required.

(*d*) Class A, grade 2 and class B buret stopcock plugs, if other than glass required.

(*e*) Number of burets per intermediate package required.

6.3 **Military use.** Items procured under this specification for military use are to be limited to the variety shown on the applicable military standards or military supply standards. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 **Supersession data.** This specification supersedes in part Federal Specification DD-V-581a; in particular, the requirements applicable to type II, burets.

6.5 **Transportation description.** Transportation description applicable to this item is:

Glassware, chemical laboratory.
Carload minimum weight 18,000 pounds,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.

subject to Rule 34, Uniform Freight Classification.

Glassware, laboratory.

Motor volume minimum weight 18,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—MU

Navy—Md

Air Force—DAY

OTHER INTERESTS:

Army—E M O

Copies of this specification may be purchased for 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:48Z
Check the source to verify that this is the current version before use.